# Presentation notes: Group 8

Ryan Merlino, April Walter, Nikolas Platon | CAI5735 Mid-Term | 5 min talk + 2 min Q&A | DRAFT: reword in your own voice

## 1. Title (about 10 sec)

Hi, we're Group 8: Ryan Merlino, April Walter and Nikolas Platon. Our project looks at diagnoses, admission type and hospital readmission for patients with diabetes.

## 2. Research question (about 15 sec)

We asked: how is the number of diagnoses related to readmission, and does that change depending on how the patient was admitted: Emergency, Urgent or Elective?

*Note: Three columns: number_diagnoses, readmitted, admission_type.*

## 3. Dataset (about 25 sec)

We used 3,000 hospital stays of patients with diabetes from 130 US hospitals, from the UCI Machine Learning Repository (Strack et al., 2014). Sixty percent were not readmitted, 31% came back after 30 days and 9% within 30 days. Emergency admissions are 57% of the stays.

*Note: No missing values.*

## 4. Analysis process (about 25 sec)

We kept the three columns we needed and made two new ones: a readmitted yes/no flag and a diagnosis group (1-4, 5-6, 7-8, 9+). We checked Emergency patients alone, then grouped by admission type and diagnosis group and sorted by readmission rate. Then we made the charts.

## 5. Most stays have 9 diagnoses (about 20 sec)

This histogram shows how many diagnoses each stay has. 1,376 of the 3,000 stays have exactly 9. That may be a recording limit in the data, so patients with more conditions could be undercounted.

*Note: We have not verified why. Say 'may', not 'is'.*

## 6. Readmitted patients have more diagnoses (about 20 sec)

Patients who were readmitted had more diagnoses on average: about 7.7, compared with 7.0 for patients who were not readmitted.

*Note: Same for both readmitted groups (after 30 days and within 30 days).*

### 7. Same pattern in every admission type (about 25 sec)

Splitting by admission type, readmitted patients still have more diagnoses in all three. Elective goes from 6.7 to 7.4 or 7.7, Urgent from 6.8 to 7.4 or 7.7, and Emergency from 7.2 to 7.7. The gap is smallest for Emergency.

### 8. Readmission rises with diagnoses (about 25 sec)

Here is the readmission rate by number of diagnoses. It climbs from about 30% at 5 diagnoses to 46% at 9. The points at 2 and 3 diagnoses are noisy because they come from small groups, 39 and 102 patients. Overall, 40% of stays ended in a readmission.

### 9. Same upward trend by admission type (about 30 sec)

This line chart splits that trend by admission type. Rates go from 24.6% with 1-4 diagnoses to 46% with 9 or more, and it levels off after about 7. Elective is lowest in every group: 34.1% overall versus 41.4% for Urgent and 41.8% for Emergency. Looking at Emergency alone shows the same rise.

### 10. Key takeaways (about 30 sec)

More diagnoses goes with a higher chance of readmission, and it levels off after about 7. That is true for all three admission types, and Elective is consistently lowest. This shows a relationship, not a cause: sicker patients may simply have more diagnoses and come back more. Also, many stays have exactly 9 diagnoses, and some subgroups are small. For example, Urgent with 1-4 diagnoses has only 74 patients.

### 11. Questions

Thank you. We're happy to take questions.

*Note: Repo: github.com/apriljwalter-lgtm/Group_8_Midterm_project*

*Total is about 4 minutes at a normal pace. If you run long, slides 5 and 6 are the easiest to shorten.*

# Key numbers (cheat sheet)

| Fact | Number |
|---|---|
| Stays / hospitals | 3,000 / 130 |
| Not readmitted / after 30 days / within 30 days | 60% / 31% / 9% |
| Admission types: Emergency / Elective / Urgent | 57% / 22% / 21% |
| Readmitted at any time | 40.0% |
| Stays with exactly 9 diagnoses | 1,376 of 3,000 |
| Avg diagnoses: not readmitted vs readmitted | 7.0 vs 7.7 |
| Readmission rate by diagnosis group (1-4 / 5-6 / 7-8 / 9+) | 24.6% / 31.9% / 44.6% / 46.0% |
| Readmission rate by type (Elective / Urgent / Emergency) | 34.1% / 41.4% / 41.8% |
| Small groups on scatter (2 and 3 diagnoses) | 39 and 102 patients |
| Urgent with 1-4 diagnoses | 74 patients |

# Likely Q&A questions (2 min)

**Does this mean more diagnoses cause readmission?**
No. It's a relationship, not a cause. Sicker patients probably have more diagnoses and are also more likely to come back.

**Why did you group the diagnoses into 1-4, 5-6, 7-8 and 9+?**
So we could compare four groups instead of a dozen separate values, many of which have very few patients.

**Why does it level off at 9?**
Many stays have exactly 9 diagnoses, which may be a recording limit in the data. We haven't verified that.

**What counts as readmitted?**
Either 'after 30 days' or 'within 30 days'. 'Not readmitted' is the 0 group.

**Why is Elective lower?**
We didn't test why. It could be that planned admissions involve less sick patients, but that's a guess.

**Did you run a statistical test?**
We compared rates and group differences with charts and summary numbers, and we did not run a formal significance test.

**What are the limits?**
A 3,000-stay sample from 1999-2008, only patients with diabetes, no severity information, and some small subgroups.